IOT PROJECT

SMART-LAB SYSTEM

MEMBERS:

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Introduction:

The rapid increase in energy consumption and security risks inside university computer laboratories highlights the need for a smart, automated management system. Traditional labs rely on manual switching of lights, computers, and door locks, which often leads to unnecessary energy waste and potential security issues when the lab is left unattended.This project introduces an IoT-based smart laboratory system capable of monitoring occupancy, controlling electrical systems, and securing the lab automatically.

Problem Statement:

University computer laboratories often suffer from unnecessary energy consumption due to lights and computer systems being left ON even when no one is present. Additionally, leaving laboratory doors unlocked increases security risks, especially after working hours.

Another major issue occurs when lecture times change unexpectedly due to circumstances beyond the instructor's control (such as schedule delays, emergency meetings, or sudden class cancellations). In such cases, students and faculty members face difficulty determining which laboratories are currently occupied or available. This leads to confusion, wasted time, and inefficient use of lab resources.

Therefore, there is a need for an automated IoT-based system that can accurately detect lab occupancy, control energy usage, secure the laboratory, and provide real-time visibility of lab availability for both students and instructors.

Objectives:

1. Detect human presence using dual PIR motion sensors.
2. Automatically control lighting, computer power, and door locks based on occupancy.
3. Reduce energy consumption when the laboratory is empty.
4. Enhance lab security through automated door-locking mechanisms.
5. Display lab occupancy status on Blynk IoT platform for real-time monitoring (occupied or empty).
6. Send automatic email notifications via Blynk when devices are turned ON (motion detected) or turned OFF (no motion).
7. Send automatic notifications via Blynk when the lab status changes.

Proposed System Description:

The SmartLab Guardian uses two PIR sensors to detect whether the lab is occupied.

The ESP32 microcontroller processes the sensor data and activates relay modules to control:

- **Lab lights**
- **Computer power outlets**
- **Electronic door locks**

When no motion is detected for a specific duration, the system automatically turns OFF all devices and locks the door.

The system integrates with Blynk IoT platform to display the lab occupancy status in real-time, showing whether the laboratory is currently occupied or empty. The system automatically sends email notifications through Blynk when:

- **Devices turned ON (Motion detected) - indicating the lab is now occupied**
- **Devices turned OFF (No motion) - indicating the lab is now empty**

This allows lab administrators to receive instant updates and monitor lab usage remotely through email notifications and the Blynk mobile app or web interface.

Hardware Components:

1. Microcontroller

ESP32 Development Board

Main processing unit

Built-in WiFi and Bluetooth

Multiple GPIO pins for sensor and relay control

Operating voltage: 3.3V

Input voltage: 5V via USB or VIN pin

2 .Sensors

PIR Motion Sensor 1 (Entrance)

HC-SR501 or similar PIR sensor

Detection range: 3-7 meters

Detection angle: ~120 degrees

Operating voltage: 5V

PIR Motion Sensor 2 (Inside Lab)

HC-SR501 or similar PIR sensor

Detection range: 3-7 meters

Detection angle: ~120 degrees

Operating voltage: 5V

3 .Relay Modules

Channel Relay Module 5V-4

Relay 1: Controls lab lights

Relay 2: Controls computer power outlets

Relay 3: Controls door lock 1

Relay 4: Controls door lock 2

Each relay rated: 10A/250VAC or 10A/30VDC

Optocoupler isolation for safety

4. Door Lock System

Electronic Door Lock 1 (Solenoid/Electric Strike Lock)

Operating voltage: 12V DC

Fail-safe or fail-secure configuration

Electronic Door Lock 2 (Solenoid/Electric Strike Lock)

Operating voltage: 12V DC

Fail-safe or fail-secure configuration

5. Power Supply

5V Power Adapter for ESP32 and relays

Output: 5V DC, 2-3A minimum

12V Power Adapter for door locks

Output: 12V DC, 2A minimum

Software Components:

1. Development Environment

Arduino IDE (v1.8.x or v2.x)

2. Required Libraries

- WiFi.h - ESP32 WiFi connectivity
- Blynk IoT platform communication
- Timer management

3. Cloud Platform

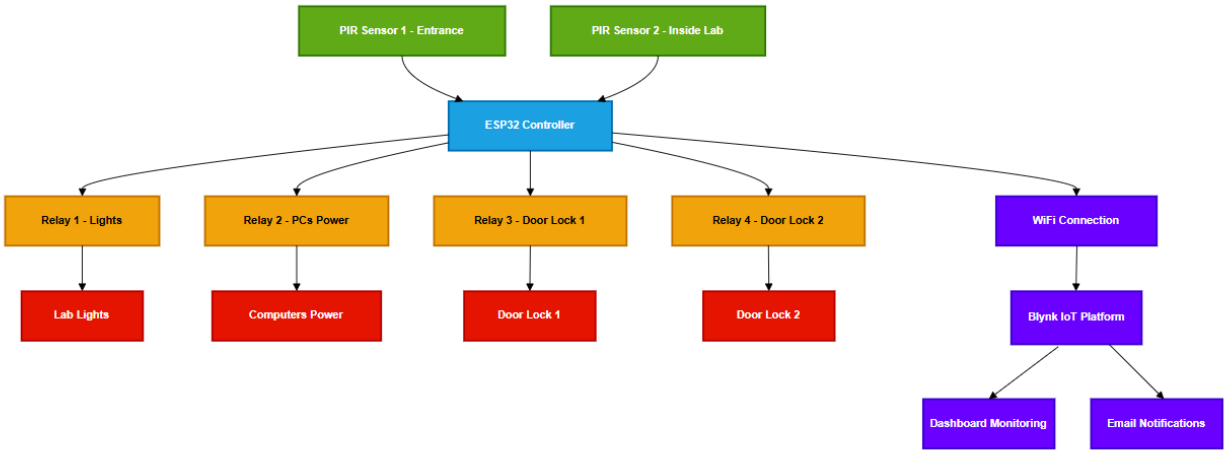
Blynk IoT Platform:

- Real-time monitoring dashboard
- Email notification service
- Mobile app (Android/iOS) and web interface

4. Programming Language

-C/C++ (Arduino framework)

Block Diagram:



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Block Diagram Description:

The SmartLab system uses two PIR motion sensors to detect the presence of students or staff inside the university computer laboratory.

The signals from both PIR sensors are sent to the ESP32 microcontroller, which serves as the main processing unit.

Based on the detected motion, the ESP32 decides whether the laboratory is occupied or empty.

When no motion is detected for a predefined duration, the ESP32 activates the relay module to automatically turn OFF the lab lights, cut the power to the computers, and lock the laboratory doors using the electric door locks.

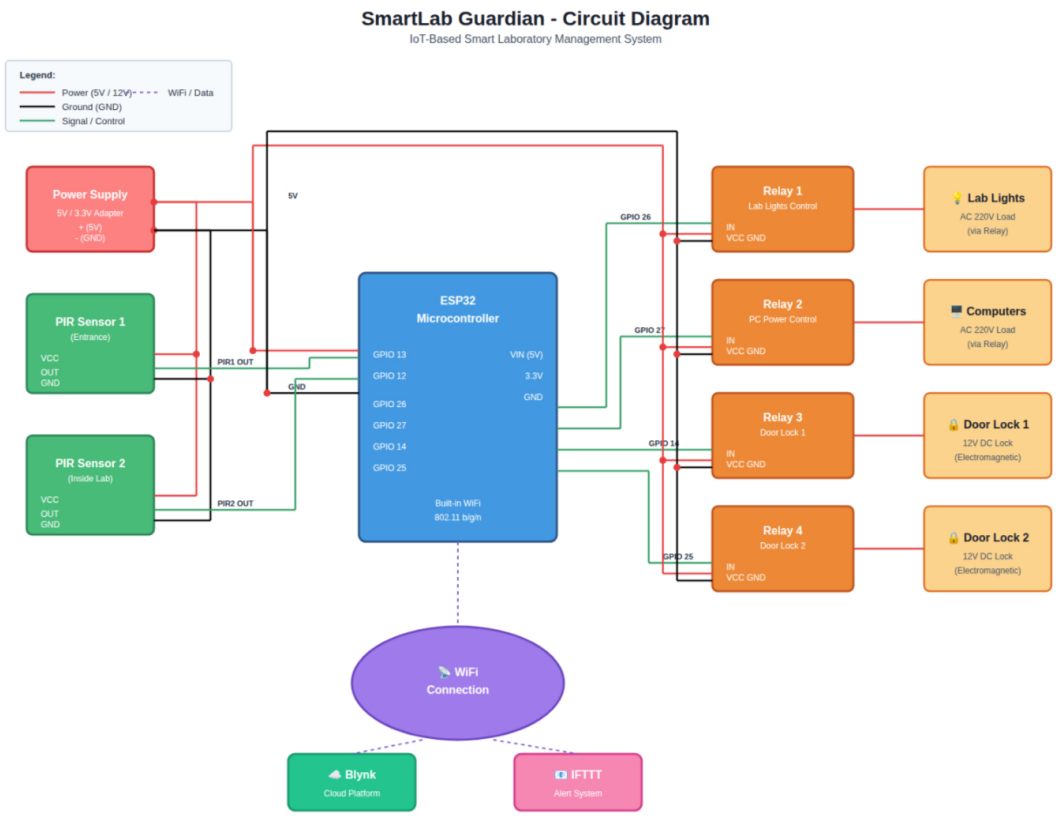
Additionally, the ESP32 sends real-time occupancy status to the Blynk IoT platform via WiFi for monitoring purposes, displaying whether the lab is currently occupied or empty.

The system also integrates with Blynk to send instant email notifications to the supervisor when:

- **Devices are turned ON (Motion detected) - indicating someone has entered the lab**
- **Devices are turned OFF (No motion) - indicating the lab has been automatically secured**

This architecture ensures an automated, energy-efficient, and secure smart laboratory environment with real-time remote monitoring capabilities.

Circuit Diagram:



Circuit Diagram Description:

This circuit diagram illustrates the SmartLab Guardian system, an IoT-based smart laboratory management solution. The system consists of an ESP32 microcontroller as the central processing unit, connected to two PIR motion sensors (one at the entrance and one inside the lab) for occupancy detection.

The ESP32 controls four relay modules through GPIO pins (GPIO 26, 27, 14, and 25), which manage the lab's electrical systems including lights, computer power outlets, and two electronic door locks. All components are powered by a 5V/3.3V power supply with proper ground connections.

The ESP32's built-in WiFi module enables communication with the Blynk IoT platform, displaying real-time lab occupancy status (occupied or empty) and sending automatic email notifications when the lab status changes. The diagram uses color-coded wiring: red for power lines, black for ground connections, green for signal/control lines, and purple dashed lines for WiFi/data communication.

This automated system reduces energy consumption by automatically controlling devices based on lab occupancy and enhances security through electronic door locks.

Conclusion

The SmartLab Guardian system successfully addresses energy waste and security issues in university computer laboratories through IoT automation. Using ESP32 and dual PIR sensors, the system automatically controls lights, computers, and door locks based on occupancy detection.

Key Results:

- **Energy Savings:** Automatic power management reduces unnecessary consumption
- **Enhanced Security:** Automated door locks secure labs when empty
- **Remote Monitoring:** Blynk and IFTTT integration enables real-time alerts and control
- **Scalable Design:** Easy deployment across multiple laboratories

This project demonstrates how IoT technology can transform traditional spaces into smart, efficient environments. The system not only saves energy and improves security but also provides a foundation for future smart campus initiatives.